



Beyond E2E Scripts: Using LLM-Proposed Scenarios Without Letting the LLM Be the Oracle

Alexander Kolpakov

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Abstract

Large language models can enlarge the scenario space explored by end-to-end tests, but they are unsafe as test oracles: they can invent impossible states, misread application semantics, and overstate weak evidence. **Bayesilisk** separates scenario proposal from verification. App-specific connectors publish a bounded search surface consisting of routes, actions, identifiers, state facts, invariants, and executable probe hooks. The **Bayesilisk** core expands that surface into candidate scenarios, rejects malformed candidates by deterministic contract validation, prioritizes valid probes with Grassmann-style attention and a Bayesian score, and verifies only evidence returned by connector execution.

This paper gives a compact formal model for that loop and reports a Cal.com case study at a fixed May 2026 revision. From connector-declared route facts and an action graph, **Bayesilisk** generated seven probes: six unknown/stale `rescheduleUid` route mutations and one bounded workflow sequence that created a booking, cancelled it, and replayed the cancelled booking UID through a public booking route. All seven returned deterministic expected-vs-observed violations. One upstream report has a targeted fix pull request that identifies the root cause and adds a regression Playwright test.

The result is not a claim that LLMs can judge application correctness; it is evidence that LLM-style proposal can be useful when the oracle remains deterministic and evidence-bound.

1 Introduction

End-to-end suites (E2E) are reliable when they fail because their assertions are explicit, executable, and repeatable. However, their weakness is coverage. A suite usually contains the workflows a team remembered to encode under the time and productivity constraints, not the combinatorial neighborhood around stale identifiers, role changes, cancelled state, direct routes, private links, feature flags, or replayed workflow tokens.

On the other hand, LLM-driven browser agents have the opposite shape. Prior work shows that language models can interleave reasoning and external actions, but realistic web-agent benchmarks still show large reliability gaps, and hallucination remains a known failure mode in generated text and reasoning [13, 14, 10]. Such agents can suggest unusual states and longer sequences, but they are unreliable when asked to decide whether the observed behavior is correct. They also cannot guarantee reliable generation of valid scenarios. Recent agentic E2E work still asks the inherently stochastic model to infer pre/postcondition oracles [12]; **Bayesilisk** instead keeps the oracle within deterministic connector-supplied bounds.

Bayesilisk [1] is built around a narrow separation: models propose where to look, but deterministic rules decide what counts as a valid scenario, and what constitutes a failure. The connector for an application exposes context and executable hooks; it does not own the scenario search strategy. The **Bayesilisk** core

generates candidate scenarios from the declared surface, validates candidates against the connector contract, ranks valid probes, asks the connector to execute them, and verifies returned evidence against explicit invariants.

This is close in spirit to property-based testing, where generators produce inputs and properties act as oracles [7]. The difference is the target: **Bayesilisk** treats a modern UI/API application as the system under test and uses browser/API probes, implemented in the artifact with Playwright, as evidence sensors [11]. It also treats LLM proposals as untrusted generator hints, not as properties and not as verdicts.

The practical value is that teams do not have to choose between narrow E2E scripts and unconstrained LLM agents. If a team can describe the app surface it already understands—actors, routes, actions, fixtures, identifiers, and rules—**Bayesilisk** can sweep the adjacent scenario space, prioritize suspicious combinations, and emit deterministic issue payloads. Our claim is deliberately bounded but useful: **Bayesilisk** does not replace domain knowledge; it turns that domain knowledge into a larger, more variable, and repeatable search procedure.

This paper makes four contributions:

1. a trust-boundary model for using LLM-proposed scenarios without using an LLM as the oracle;
2. a mathematical formulation of connector contracts, scenario validation, Grassmann-style attention, Bayesian ranking, and deterministic evidence verification;
3. a Cal.com artifact showing seven generated probes, seven observed expected-vs-observed violations, reproducible issue payloads, and an upstream targeted fix pull request for one reported issue; and
4. an explicit account of scope and generalization: connectors are app-specific by design, action-sequence breadth is governed by the connector-declared graph, and the Cal.com comparison runs validate the architectural separation from E2E-only, Playwright-only, weak-model, and LLM-agent workflows.

2 Claim Audit

There are four important questions to be answered about **Bayesilisk**, including its architecture and use cases.

Who generates scenarios? The connector declares the app-specific search surface: routes, actions, identifiers, state facts, invariants, and valid mutation or action schemas. **Bayesilisk** generates concrete candidate scenarios from that declared surface, optionally using LLM proposals as additional candidate ideas. The connector does not enumerate final tests.

Who executes scenarios? The connector executes validated probe requests through the real application, local fixtures, Playwright, API calls, or a test harness. It returns observed evidence such as semantic status, URL, trace, screenshot, state transition, or execution error.

Who decides pass or fail? **Bayesilisk** decides pass/fail by applying deterministic invariants to evidence returned by the connector. Neither the LLM nor the connector is the oracle.

What external validation exists? The Cal.com artifact produced issue-ready findings. One public issue, #29399 [2], has an upstream contributor fix pull request, #29400 [5], that identifies a missing null guard in `processReschedule` and adds a regression Playwright test. A separate workflow issue thread, #29407 [3], remains open at capture time.

3 Formal Model

Let an application connector publish a contract

$$C = (R, A, F, I, M, G),$$

where R is a set of route patterns, A is a set of connector actions, F is a set of observable state facts and identifiers, I is a set of deterministic invariants, M is a set of parameter mutation schemas, and G is an action graph. The connector is app-specific; the **Bayesilisk** core is a generic function over C .

3.1 Scenario Generation and Validation

From connector-declared route facts and mutation schemas, **Bayesilisk** constructs a route-mutation candidate of the form

$$s = (r, a, i, p, m),$$

where $r \in R$ is a declared route pattern, $a \in A$ is a declared connector action, $i \in I$ is a declared invariant, p is a declared route parameter, and $m \in M(p)$ is an allowed mutation for that parameter, such as unknown, stale, foreign, missing, or duplicate. The connector does not enumerate this final candidate. It declares the searchable surface and the allowed mutation schema; **Bayesilisk** performs the expansion, deduplication, validation, and ranking.

An action-sequence candidate is generated from an action graph

$$G = (A, \text{req}, \text{prod}),$$

where each action has required tokens and produced tokens. A sequence

$$\sigma = (a_1, \dots, a_k)$$

is valid for a terminal goal action a_k only if

$$\forall t \in \{1, \dots, k\} : \text{req}(a_t) \subseteq \bigcup_{j < t} \text{prod}(a_j), \quad k \leq d_{\max}.$$

Thus a connector can expose actions such as create booking, cancel booking, and open booking route, while **Bayesilisk** composes the bounded workflow. The connector still owns concrete execution.

Validation is deterministic:

$$V_C(s) = \begin{cases} 1 & \text{if every referenced route, action, parameter, fact, and invariant is declared in } C, \\ 0 & \text{otherwise.} \end{cases}$$

Only candidates with $V_C(s) = 1$ can become executable probes. LLM output enters the same validation gate. Unknown target planes, unknown ids, duplicate proposals, non-string fields, malformed JSON, impossible counts, and unsupported actions are rejected as proposer telemetry rather than reported as product failures.

3.2 Grassmann Attention

The attention layer is a prioritizer, not an oracle. Conceptually, invariant descriptions and observations define local subspaces in a feature space. For an invariant plane P_i and context vector q , a Grassmannian proximity score can be written as

$$\gamma_i(q) = \|\Pi_{P_i} q\|_2^2,$$

where Π_{P_i} projects q onto the invariant plane. This follows the standard geometric view of subspaces on Grassmann manifolds [8].

However, $\gamma_i(q)$ is only an ideal geometric score. The current implementation does not compute this projection directly. Instead, it uses an interpretable bounded anchor-plane proxy, α_i , whose novelty term can include embedding similarity to the same invariant-plane anchors. Thus α_i is the operational attention score used for ranking planes in the current artifact, while γ_i describes the intended Grassmannian interpretation. For each invariant i , **Bayesilisk** computes

$$\alpha_i = \text{clip}_{[0,1]}(w_f f_i + w_u u_i + w_s \sigma_i + w_p p_i + w_n n_i - d_i),$$

with current weights

$$(w_f, w_u, w_s, w_p, w_n) = (0.45, 0.25, 0.15, 0.10, 0.05).$$

The variables are computed directly from context facts. The notation is: P_i is the invariant plane; q is the context vector; Π_{P_i} is projection onto P_i ; $\gamma_i(q)$ is ideal Grassmann proximity; α_i is the operational attention score; T_i is the number of facts whose `invariantId` is i ; F_i is the number of those facts that fail; N_i^{PW} is the

number of those facts whose source is Playwright; K_i is the keyword-hit count; E_i is embedding similarity; σ_i is configured invariant sensitivity; and d_i is fixed/muted decay. The normalized feature names are: f_i for failure density, u_i for untestedness, p_i for Playwright evidence, and n_i for novelty.

The current artifact uses three saturation thresholds:

$$\tau_T = 4, \quad \tau_P = 3, \quad \tau_K = 6.$$

They are calibration constants, not oracle rules. τ_T means that after four observations a plane is no longer treated as materially untested; τ_P means that three Playwright observations saturate the browser-evidence term; and τ_K means that six keyword hits saturate lexical novelty. These values are current implementation defaults chosen to make attention saturate quickly on small connector artifacts; they should be treated as tunable ranking parameters and ablated in larger evaluations. Changing these thresholds changes prioritization pressure only; it does not change candidate validity or pass/fail verification. Then

$$f_i = \begin{cases} F_i/T_i & T_i > 0, \\ 0 & T_i = 0, \end{cases} \quad u_i = \begin{cases} 1 & T_i = 0, \\ \max(0, 1 - \min(T_i, \tau_T)/\tau_T) & T_i > 0, \end{cases} \quad p_i = \min(1, N_i^{PW}/\tau_P).$$

In the current artifact, the sensitivity σ_i is read from the `INVARIANT_SENSITIVITY` table maintained by the `Bayesilisk` authors in the core configuration. It encodes how sensitive an invariant plane is considered for search prioritization. If an invariant id is absent from that table, as may be the case for connector-supplied external invariants, the implementation uses the neutral default 0.5. The novelty term is

$$n_i = \max(\min(1, K_i/\tau_K), E_i),$$

where K_i is the context keyword-hit count for invariant i , and $E_i \in [0, 1]$ is optional embedding similarity to the invariant anchor plane (0 if embeddings are disabled or unavailable). The decay d_i is 0.12 for fixed or regression-watch invariant ids, 0.20 for muted invariant ids, and 0 otherwise, using the larger value if both apply. The largest α_i values select planes for exploration. They do not determine whether a behavior is correct.

3.3 Bayesian Ranking

Bayesian ranking runs only after a deterministic invariant result exists. It is used to order findings in the report, assign posterior-mode labels, and decide which issue candidates or probes appear first. It is not used to create a failure.

For catalog and generated verifier scenarios, the ranking inputs come from two places: invariant metadata and connector/context observations. In the current implementation the invariant metadata is the `Invariant` object:

$$I_i = (\text{id}, \text{expected}, \pi_i, L_i^{\text{fail}}, L_i^{\text{pass}}, \text{difficulty}, \phi_i).$$

The `Bayesilisk`-authored catalog stores these objects in `bayesilisk/invariants.py`; report output serializes the same likelihood metadata as `failLikelihood` and `passLikelihood`. Thus π_i , L_i^{fail} , and L_i^{pass} are not inferred during a run. They are prior/ranking constants supplied by the invariant author with the invariant definition; they are not generated by the LLM and are not derived from connector execution. For connector-defined invariants, the same role is played by the connector’s declared invariant metadata.

The run-time context adjustment is computed from observation tags and explicit context fields. For finding fingerprint x , invariant i , and scenario s , the implementation computes

$$\Delta_i = -0.28 \mathbf{1}_{x \in \text{Fixed}} + 0.18 \mathbf{1}_{x \in \text{Confirmed}} - 0.45 \mathbf{1}_{x \in \text{Muted}} + A_i + B_s,$$

where x is the finding fingerprint, A_i is an explicit context adjustment for invariant i , and B_s is an explicit context adjustment for scenario s ; missing explicit adjustments are zero. The sets `Fixed`, `Confirmed`, and `Muted` are populated from `fixedFingerprints`, `confirmedFingerprints`, and `mutedFingerprints` in the context/observation file, plus fingerprints extracted from that context. A_i is read from the context field `priorAdjustments`; B_s is read from `scenarioAdjustments`. If these fields are absent or malformed, their contribution is zero.

The likelihood used in the posterior is selected after deterministic verification. For a concrete scenario s , **Bayesilisk** merges its fragments into facts F_s and calls the invariant evaluator:

$$\phi_i(F_s) \rightarrow (b_i, o_i),$$

where b_i is the Boolean evaluator result and o_i is the evaluator’s observation text. The reported deterministic result is

$$r_i = \begin{cases} \text{pass} & \text{if } b_i = \text{true}, \\ \text{fail} & \text{if } b_i = \text{false}. \end{cases}$$

Only then does **Bayesilisk** choose the scalar likelihood used for ranking:

$$\ell_i = \begin{cases} L_i^{\text{pass}} & \text{if } r_i = \text{pass}, \\ L_i^{\text{fail}} & \text{if } r_i = \text{fail}. \end{cases}$$

So “computed” here means deterministic branch selection from invariant metadata after the evaluator has already returned pass/fail. There is no learned likelihood estimator in the current artifact, and the likelihood does not decide whether the scenario failed. It only changes the ordering of already evaluated findings. The adjusted prior is

$$\tilde{\pi}_i = \text{clip}_{[0.05, 0.95]}(\pi_i + \Delta_i).$$

The posterior risk score is

$$\rho_i = \frac{\tilde{\pi}_i \ell_i}{\tilde{\pi}_i \ell_i + (1 - \tilde{\pi}_i)(1 - \ell_i)}.$$

This is a ranking score in the Bayesian sense [4]. Pass/fail is produced only by the invariant evaluator.

Direct connector-observed findings use the same general report pipeline but a different scoring path. If connector evidence already contains a deterministic mismatch between expected and observed status, the finding is evidence-bearing before Bayesian ranking. **Bayesilisk** records it as a context-observed breakage, assigns a high report-ordering score, and marks it issue-ready. This path is connector-generic; Cal.com is the current case study using it, not a special rule in the core.

3.4 Evidence Verification

Execution is a connector function

$$X_C(s) \rightarrow e,$$

where e is observed evidence. For the Cal.com connector, the core evidence fields are **expectedStatus**, **observedStatus**, route, target URL, invariant id, and source context. The verifier applies

$$\phi_i(e) = \begin{cases} \text{fail} & \text{if } e.\text{observedStatus} \neq e.\text{expectedStatus}, \\ \text{pass} & \text{if } e.\text{observedStatus} = e.\text{expectedStatus}, \\ \text{invalid} & \text{if required evidence is absent or malformed.} \end{cases}$$

Only fail with evidence becomes an issue-ready finding. **invalid** is a connector/proposer diagnostic.

4 Implementation Boundary

Bayesilisk implements three distinct ledgers. The attention ledger records which invariant planes should be explored. The model-proposal ledger records accepted and rejected model proposals. The verification ledger records deterministic invariant outcomes, fingerprints, issue readiness, and risk scores. Only the verification ledger contains pass/fail results.

This boundary matters for connector design. A connector should expose a small vocabulary of actions and facts. It should not filter away uncomfortable cases or decide that a 200 response is acceptable. The finite-state verifier owns the ruleset, which is supplied by context and schema, not hardcoded for a particular application. App-specific code is isolated to connector execution.

5 Cal.com Case Study

We evaluated the current connector architecture on a local Cal.com checkout [6]. The tested repository and revision were:

- Repository: <https://github.com/calcom/cal.com>
- Commit: 180ede28f0bddf2738933a6e60a8e80f6116d7da
- Commit date: 2026-05-14 19:30:21 +0000
- Commit subject: font-stack fallback fix for non-Latin script support (#29346)
- Bayesilisk seed: 150

The connector supplied three route facts for booking pages with `rescheduleUid` behavior and one action graph for a longer booking workflow. Bayesilisk generated six route-mutation probes and one workflow-sequence probe. The sequence was:

`create-booking → cancel-booking → open-public-booking-route.`

Artifact	Path
Connector code	<code>examples/calcom/bayesilisk-probes.e2e.ts</code>
Route context	<code>examples/calcom/source-context.json</code>
Generated route proposals	<code>examples/calcom/generated-proposals.json</code>
Sequence context	<code>examples/calcom/sequence-source-context.json</code>
Generated sequence proposal	<code>examples/calcom/generated-sequence-proposals.json</code>
Execution evidence	<code>examples/calcom/execution-context.json</code>
Bayesilisk report	<code>examples/calcom/reports/report.json</code>
Issue payloads	<code>examples/calcom/reports/issue-payloads.json</code>
Upstream outcomes	<code>examples/calcom/upstream-outcomes.md</code>

Table 1: Checked-in Cal.com artifact set.

Area	Mutation	Expected	Observed
Public booking page	unknown id	404	200
Public booking page	stale id	404	200
Private booking link	unknown id	404	200
Private booking link	stale id	404	200
Dynamic booking page	unknown id	404	200
Dynamic booking page	stale id	404	200
Cancelled booking replay	cancelled uid	409	200

Table 2: Verified Cal.com connector observations.

All seven probes executed against local Playwright fixtures without unexpected Playwright test failures [11]. All seven returned evidence that violated the connector-declared invariant. The route mutations expected semantic status 404 and observed 200. The cancelled-booking replay expected semantic status 409 and observed 200. Bayesilisk classified the findings as `breakage.context-observed` and produced issue-ready payloads.

6 Comparison Runs

The evaluation includes small comparison runs that clarify the role of Bayesilisk. These are not a broad benchmark; they are local controls around the Cal.com artifact.

Workflow	Observation
Nearby E2E	14 nearby specs ran without unexpected failures; the existing suite did not cover the generated stale/unknown states.
Playwright-only probing	11 manual observations produced 4 failures; manual probes exposed only a subset of the bug family.
Bayesilisk route proposals	6 generated route proposals produced 6 findings across declared route variants.
Bayesilisk sequence proposal	1 generated workflow sequence produced 1 cancelled-booking replay finding.
Weak local model proposals	3 raw proposals produced 0 accepted probes because verifier validation rejected malformed candidates.
LLM-agent browser run	10 trials produced 3 oracle mismatches before deterministic verification.

Table 3: Cal.com comparison summary.

The comparison supports a bounded claim. Manual Playwright probes can find some instances when a human already knows where to look. **Bayesilisk** broadens coverage by expanding declared route facts and action graphs into concrete probes. The weak model result is also useful: invalid model proposals were contained by schema and verifier validation rather than converted into product bugs. The LLM-agent browser run illustrates why the model should not be the oracle.

7 Upstream Validation

Two upstream issue threads were opened from the Cal.com evidence. The first, #29399 [2], reported that an unknown `rescheduleUid` opened a normal booking flow instead of returning 404. An upstream contributor opened PR #29400 [5]. The PR identifies the root cause as a missing null guard in `processReschedule`, changes the server-side behavior to return `notFound`, and adds a Playwright regression test. The private-link variant was added to the same issue thread to avoid duplicate reporting. A separate issue, #29407 [3], reports the cancelled booking UID replay sequence and remains open at capture time.

Finding group	Upstream response
Public unknown/stale <code>rescheduleUid</code>	Issue #29399 with targeted fix PR #29400: null guard and regression test
Private-link <code>rescheduleUid</code>	Added to issue #29399 as a related comment to avoid duplicate issue noise
Cancelled booking UID replay	Separate state-transition report in issue #29407; open at capture time

Table 4: Upstream response to Bayesilisk-generated findings.

This validation is practical rather than statistical. A fix PR does not prove the entire methodology, but it does show that at least one generated finding was actionable enough for an upstream contributor to diagnose and patch.

8 Scope and Generalization

Bayesilisk is intended to be adopted through app-specific connectors and app-agnostic verification. The following points describe that engineering boundary and the evidence needed to broaden the empirical case; they are not exceptions to the architecture.

Application scope. The current public practical validation is Cal.com. The result does not depend on Cal.com-specific verifier code: the core consumes a connector contract and generated probes through the same app-agnostic interfaces used elsewhere in **Bayesilisk**. Additional application artifacts would

strengthen empirical breadth, but the required adaptation is the expected one: each app team writes a connector that exposes its own routes, actions, state facts, fixtures, and invariants.

Connector specificity. Connector-specific code is not a weakness of the architecture; it is the integration boundary. A team that knows the app specification can declare actors, routes, action hooks, identifiers, and invariants without modifying the **Bayesilisk** core. Connector quality still matters: the broader and more accurate the declared surface, the broader the generated probe space.

Bounded workflow generation. Sequence generation is intentionally bounded by the connector-declared action graph and depth. That bound can still cover a large workflow space when the connector exposes a broad action vocabulary and meaningful state tokens. The design avoids arbitrary free-form browser wandering; **Bayesilisk** composes declared actions and the connector executes them.

Oracle scope. The deterministic oracle is only as good as the supplied ruleset. **Bayesilisk** prevents the LLM from being the oracle; it does not remove the need for correct invariants.

Baseline interpretation. The Cal.com comparison runs validate the architectural difference: nearby E2E did not cover the generated states, manual Playwright probes were narrower, weak local model output was rejected by validation, and browser-driving LLM trials showed oracle instability. A larger multi-app study would quantify effect size across projects and models; it is not required to establish the core separation demonstrated here.

9 Future Work

Two next steps would make the ranking layer more mathematically grounded while preserving the same trust boundary: attention and probability may guide search, but they must not become the oracle.

Actual Grassmann attention. The current implementation uses α_i , a bounded anchor-plane proxy for the ideal Grassmannian score $\gamma_i(q)$. A fuller version should construct an explicit feature vector q for each connector context and an orthonormal basis U_i for each invariant subspace P_i . The operational attention score could then be the projection energy

$$\gamma_i(q) = \|U_i^\top q\|_2^2,$$

or a related Grassmann distance based on principal angles between the context subspace and invariant subspace. The basis vectors can be built from invariant text, connector-declared routes/actions, nearby tests, execution evidence, and validated historical findings. This would replace the current hand-weighted proxy as the main exploration router, while leaving deterministic verification unchanged [8].

Calibrated likelihoods. The current artifact uses invariant-level ranking constants L_i^{fail} and L_i^{pass} . A stronger version should estimate likelihoods from validation data. Let Z_i mean that invariant i is genuinely violated, and let $r_i \in \{\text{pass}, \text{fail}\}$ be the deterministic verifier result. For a failing result, the calibrated posterior should use

$$P(Z_i \mid r_i = \text{fail}) = \frac{P(Z_i)P(r_i = \text{fail} \mid Z_i)}{P(Z_i)P(r_i = \text{fail} \mid Z_i) + P(\neg Z_i)P(r_i = \text{fail} \mid \neg Z_i)}.$$

The terms $P(r_i = \text{fail} \mid Z_i)$ and $P(r_i = \text{fail} \mid \neg Z_i)$ are the true-positive and false-positive behavior of that verifier/probe family. They can be estimated from historical Bayesilisk findings, seeded faults, negative controls, connector evidence quality, invariant class, and human validation signals such as upstream issue acceptance, fix pull requests, and regression tests. Passing controls should be calibrated analogously with $P(r_i = \text{pass} \mid Z_i)$ and $P(r_i = \text{pass} \mid \neg Z_i)$. This would turn the current hand-calibrated ranking constants into empirical reliability estimates without changing the rule that pass/fail is decided only by deterministic invariants [4, 9].

10 Conclusion

Bayesilisk treats LLMs and coding agents as scenario proposers, not as test oracles. The mathematical boundary is simple: candidate generation and attention may be probabilistic or heuristic, but validation and verification are deterministic functions over a connector-declared contract and connector-returned evidence.

The Cal.com case study demonstrates that this separation can produce useful real-app findings: seven generated probes, seven observed invariant violations, issue-ready payloads, and a targeted upstream fix PR for one reported issue. The adoption path is incremental: write the connector around existing app fixtures, let **Bayesilisk** expand the scenario space, and keep the oracle deterministic. The next research step is to repeat this artifact discipline across additional applications while keeping the **Bayesilisk** core app-agnostic.

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